Swayam Gupta

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SUMMARY

AI engineer with full-stack range, building RAG pipelines and multi-agent orchestration end to end. Next.js and TypeScript on the frontend, Python and Node on the backend. Built a medical-domain RAG system with Reciprocal Rank Fusion retrieval and a custom evaluation harness, and shipped a real-time multi-agent AI platform with 50+ users in week one.

EDUCATION

Jaypee Institute of Information Technology **Noida, Uttar Pradesh**
Bachelor of Technology, Computer Science and Engineering | CGPA: 7.84/10 *July 2027*

PROJECTS

Medical/Academic PDF RAG System (Python / FastAPI) - https://github.com/SwayamGupta12345/Medi_back
Backend / AI Infra | FastAPI, Pinecone, Gemini, LangChain, CrewAI, PyMuPDF

- Built a document QA system over medical PDFs, starting with a custom ingestion pipeline that strips watermark and boilerplate text using font size, rotation, and opacity heuristics, automatic title extraction from PDF metadata and font-based heading detection.
- Implemented Reciprocal Rank Fusion for retrieval, combining dense vector search over 3072-dimensional Gemini embeddings with keyword-overlap ranking instead of relying on cosine similarity alone.
- Added batched, rate-limited embedding calls with exponential backoff after hitting live Gemini API quota limits during bulk PDF uploads, along with embedding-dimension validation at both ingest and query time.
- Built a retrieval evaluation harness measuring precision@k, recall@k, MRR, and NDCG against a 40-query labeled set, achieving 95% MRR and 100% recall@5, to validate hybrid retrieval quality instead of relying on spot checks.

Mirai - Real-Time Multi-Agent AI Collaboration Platform (Python + GenAI) - https://mirai-pi.vercel.app | Open Source
Full-Stack / AI | Next.js (TypeScript), Node.js, Python (FastAPI), WebSockets

- Architected and deployed a decoupled full-stack system (Next.js on Vercel, FastAPI on Render) with OAuth authentication, maintaining stable multi-service performance under concurrent user load.
- Built a real-time collaborative AI chat system using WebSockets with shared context, live message synchronization, and streaming responses, ensuring sub-second latency communication and consistent message edit and delete behavior during concurrent usage.
- Designed and built the backend API for a multi-agent platform using FastAPI, coordinating specialized agents for reasoning, coding, and writing; onboarded 50+ users in week one. Open-sourced the project on GitHub and maintaining active issues.

Jaypee Learning Hub - Academic Resource Management Platform (Full-Stack) - https://jaypeelearninghub.great-site.net
Web Development

- Built and scaled an academic platform to 1000+ active users within 4 months, managing 1200+ resources.
- Architected the entire data model and backend logic independently, later refactoring database schemas and optimizing SQL queries to improve load times by 30%, data consistency, and overall system responsiveness.
- Structured and designed pages using semantic HTML and responsive CSS supporting screen widths down to 240px, from rough wireframe sketches, built entirely in vanilla JavaScript, CSS, and PHP with no frameworks.

WORK EXPERIENCE

Style My Space (AI Research & Product Intern) - **Remote** *Jan 2026- March 2026*

- Sole-engineered a full-stack AI evaluation platform (Next.js, MongoDB, Cloudflare R2) from scratch, enabling multi-model image generation, structured comparison, and rating workflows adopted directly into team operations.
- Architected and executed large-scale prompt pipelines across multiple generative models and 50+ prompt variations, driving data-driven model selection evaluated across prompt adherence, output clarity, structural correctness, and consistency.
- Engineered a comparison dashboard handling 1,000+ generated assets, implementing rate limiting and optimized bulk queries to prevent database overload; added client-side pagination and viewport-based lazy loading for smooth rendering.
- Established idempotent image handling workflows with standardized naming conventions across all generated assets, ensuring clean version control, zero duplication, and reliable retrieval, reducing asset handling overhead.
- Built client-side generation and prompting interfaces enabling rapid iteration across prompt variations, directly accelerating model evaluation workflows and cutting testing turnaround time.

SKILLS

Tools and Platforms: Git, GitHub, Docker, DockerHub, Linux (CLI, Bash), Pinecone, AWS
Data & Analytics: SQL (MySQL), NumPy, Pandas, Scikit-learn, Data Preprocessing, Model Training, RAG, LangChain, CrewAI
Programming Languages: Python, TypeScript, JavaScript, C/C++, PHP
Web & Backend Technologies: HTML, CSS, DOM, Node.js, React, Express.js, FastAPI, WebSockets, MongoDB, Next.js
Soft Skills: Agile Project Management, Cross-functional Collaboration, Technical Leadership, Public Speaking

ACHIEVEMENTS

- Solved 660+ problems on LeetCode with a contest rating of 1,663 - https://leetcode.com/u/Swayam_Gupta .
- AWS Cloud Practitioner coursework completed. Certificate
- McKinsey Forward Program - strategic thinking, agile problem-solving, leadership, and communication. Certificate Badge
- Technical Head, AI/ML Club - Organized a club-level hackathon and mentored peers on ML and system design.
- Completed Krish Naik's Data Science, ML, DL & NLP Bootcamp on Udemy. Certificate